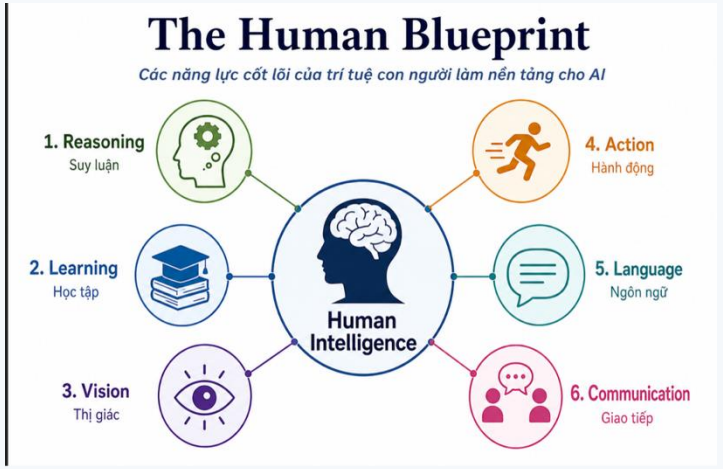
INTRODUCTION TO

ARTIFICIAL INTELLIGENCE

From questions about intelligence to data, machine learning, and responsible use

CHAPTER 04 DIGITAL FOUNDATIONS



6 capabilities • 1 big question

Ho Chi Minh City University of Technology • Introductory lecture

Learning objectives

After this lesson, students will be able to…

01 Explain AI

Describe what AI is — and what it is not.

02 Recognize applications

Distinguish vision, language, generative AI, and robotics.

03 Describe machine learning

Explain the roles of data, models, and evaluation.

04 Use AI responsibly

Verify outputs, protect data, and preserve human oversight.

Guiding question: what turns a rule-following computer into a system that can learn?

Lesson roadmap

Concepts → mechanisms → applications → responsibility

01

What is AI?

Intelligence, history, capabilities

→

02

What can AI do?

Vision, NLP, generation, robotics

→

03

How does AI learn?

Data, functions, three learning paradigms

→

04

How should we use AI?

Cases, risks, verification

WARM-UP

Where have you used AI today — even when you could not see it?

01

What is AI?

Where did AI begin, how does it differ from traditional programming, and what capabilities make up ‘intelligence’?

SECTION BREAK

AI: a system with capabilities, not an ‘artificial brain’

AI is the field of building systems that can perform tasks commonly associated with human intelligence.

✓ A useful view

Pattern recognition, learning from data, reasoning, communication, and decision support.

≠ A misleading view

Modern AI is not a complete copy of a human; its capabilities are usually narrow and data-dependent.



Humans can ‘look and know.’ Machines need data + a model + an objective.

A system that recognizes cats does not automatically understand the world as humans do.

From rule-based programming to learning from data

When exceptions cannot be fully enumerated, a model learns patterns

TRADITIONAL PROGRAMMING

- Humans write rules: IF temperature < 18°C → turn on heating
- Best for: clear tasks and stable conditions
- Limitation: many exceptions → unwieldy rules

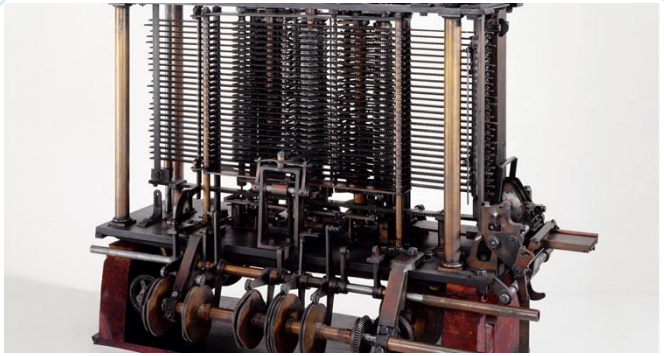
MACHINE LEARNING

- Humans provide: data + an objective
- The model learns: patterns mapping inputs → outputs
- Trade-off: representative data and testing are required

RULES → DATA → MODEL → PREDICTION

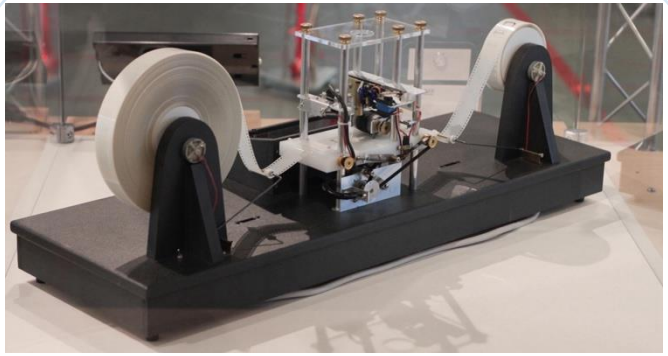
Three milestones that paved the way for AI

Computation → theoretical foundations → a research field



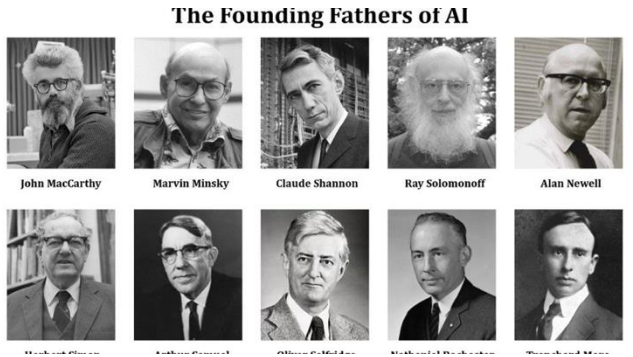
1837 **Analytical Engine**

A mechanical computer design with processing, memory, and control.



1936 **Turing Machine**

A formal model of symbol manipulation and computability.



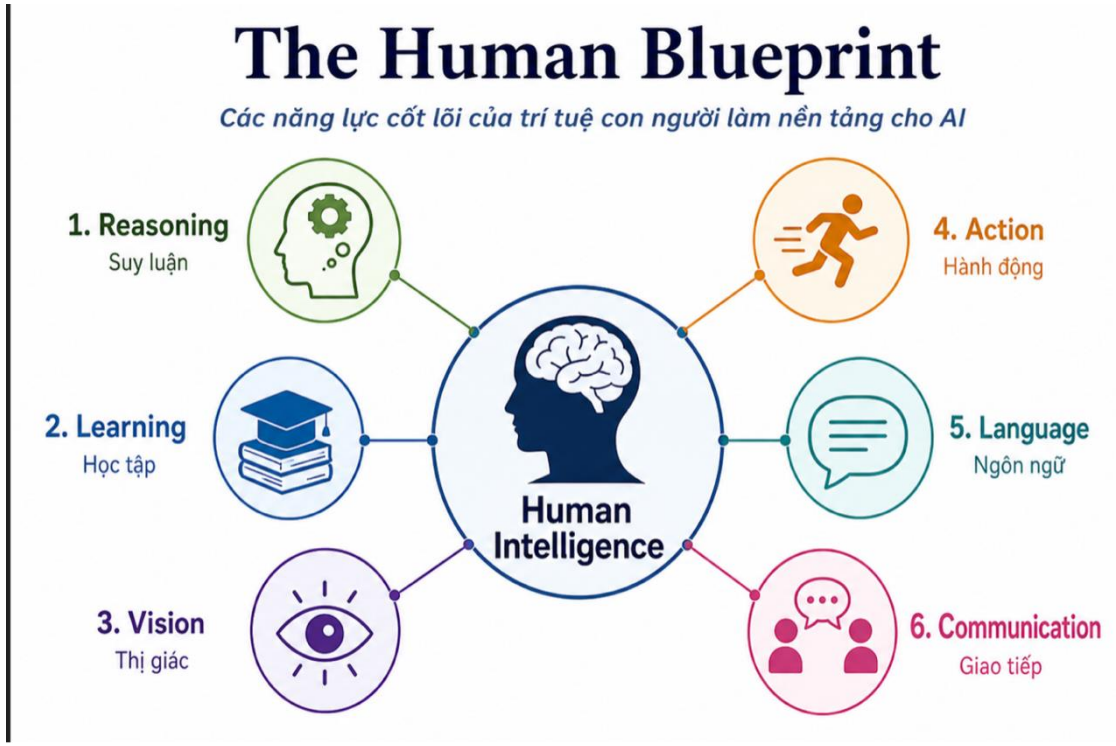
1956 **Dartmouth**

Artificial Intelligence was established as a distinct research field.

Historical lesson: bold ambition drove progress, but real-world intelligence proved more complex than early researchers expected.

A ‘blueprint’ of intelligence

AI is not a single skill



Reasoning • Learning • Vision • Action • Language • Communication

- Strong today: recognition, content generation, language processing
- Advancing quickly: robotics and action in physical environments
- Still challenging: long, consistent, verifiable reasoning

Fluent ≠ correct

A convincing output can still hallucinate.

02

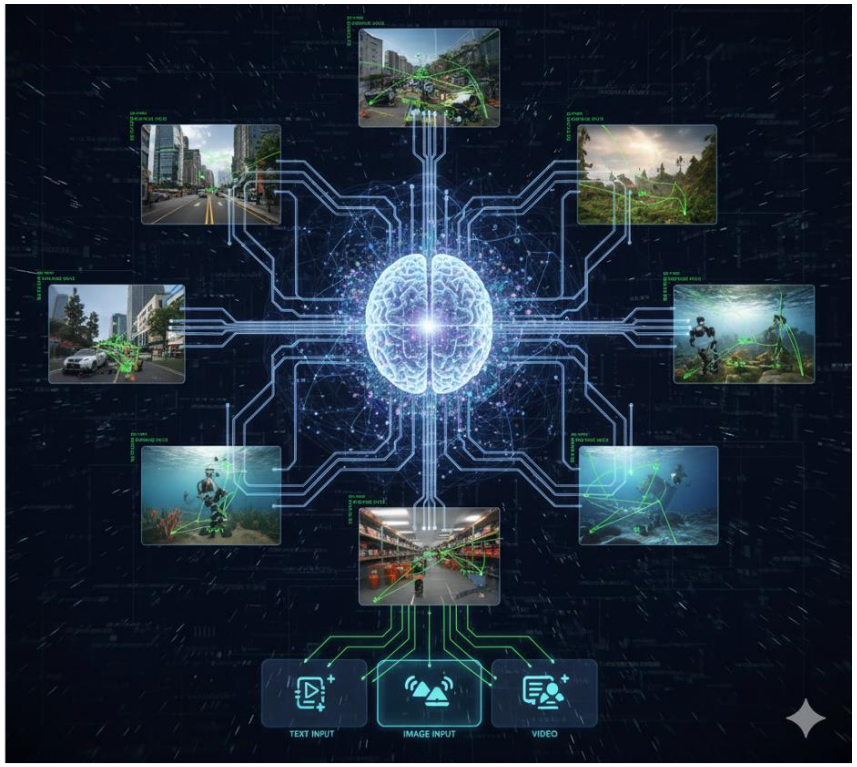
What can AI do?

What capabilities power smart cameras, chatbots, generative content, and robots?

SECTION BREAK

APPLICATION

Computer vision: from ‘seeing’ to ‘understanding’



One image → multiple levels of understanding

01 Classification

What is in the image?

02 Detection

Where is the object?

03 Segmentation

Which pixels belong to the object?

Healthcare • agriculture • transportation • manufacturing

Natural language processing: context determines meaning

“Great, another assignment on Sunday night.”

Positive or sarcastic?

- Difficult because of: ambiguity, context, culture, slang
- Requires: language patterns + world knowledge

01 Machine translation

Translation requires phrases and context, not word substitution.

02 Summarization & QA

Extract key ideas and respond to a request.

03 Sentiment analysis

Classify feedback while accounting for sarcasm.

04 Speech recognition

Turn an audio signal into usable content.

Generative AI: from predicting labels to creating new content



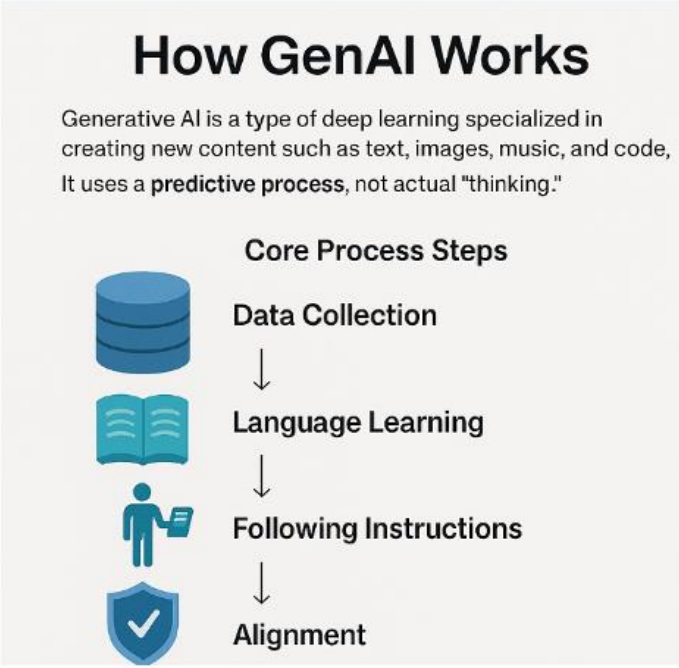
Separate categories vs. Create new data points

- Input: a prompt — objective, context, constraints
- Output: text, images, audio, video, code
- Strength: lowers the cost of producing a first draft
- Condition: humans verify correctness and sources

FIRST DRAFT ≠ FINAL ANSWER

Generative AI operates through probability and context

Natural responses do not imply human-like awareness



The model learns statistical patterns and predicts a plausible continuation for the context.

01 Data

Learn patterns from large datasets

02 Attention

Weigh relationships across the context

03 Generate output

Predict the next token or unit

04 Verification

Use sources, tools, and people to validate

The Transformer (2017) enables long-range relationships and efficient large-scale training.

AI beyond the screen: robots and edge devices

ROBOTS / AUTONOMOUS VEHICLES

- Sense: cameras, radar, lidar, other sensors
- Predict: objects and what may happen next
- Act: control under noise and latency
- Risk: errors can cause physical consequences

THE REAL WORLD = imperfect observations + physics + safety



EDGE AI

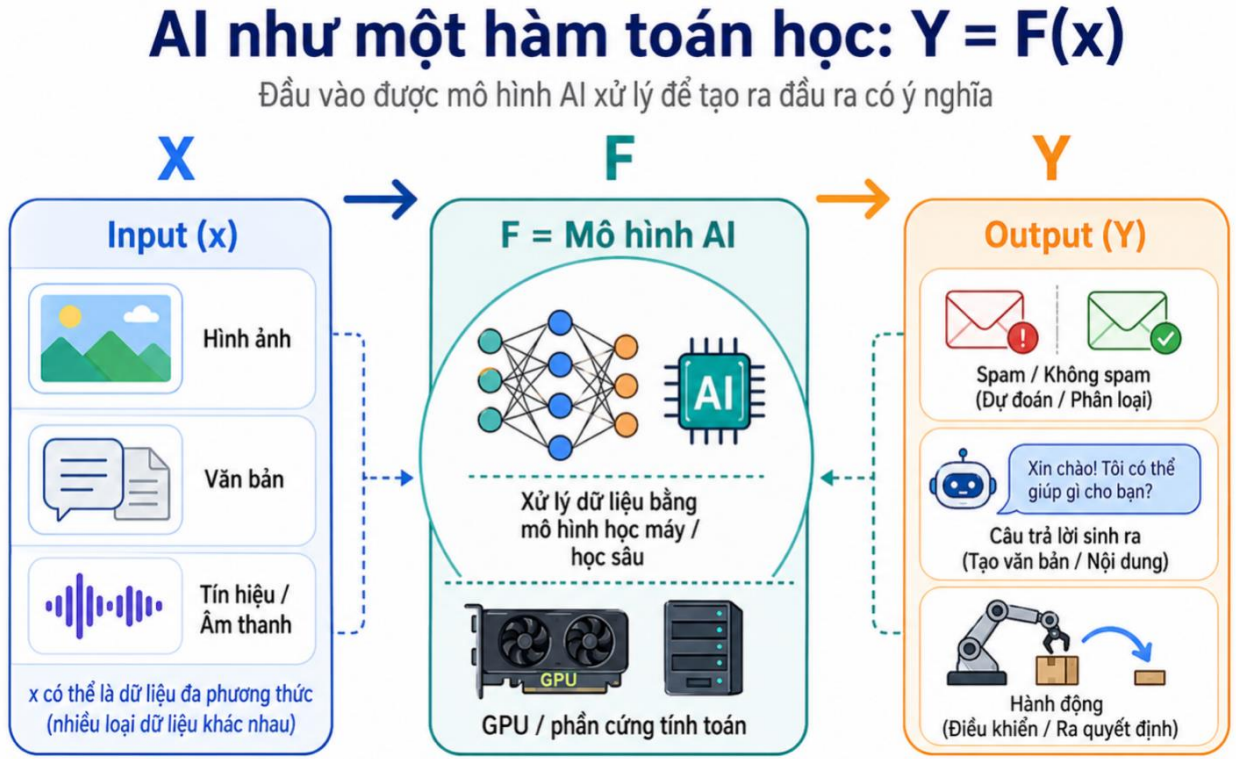
Process on the camera/device → lower latency, less bandwidth, better privacy; in exchange, compute is limited.

03 How does AI learn?

How does a model turn data into predictions, and how do learning paradigms differ?

SECTION BREAK

AI as a mathematical function: Y = F(X)

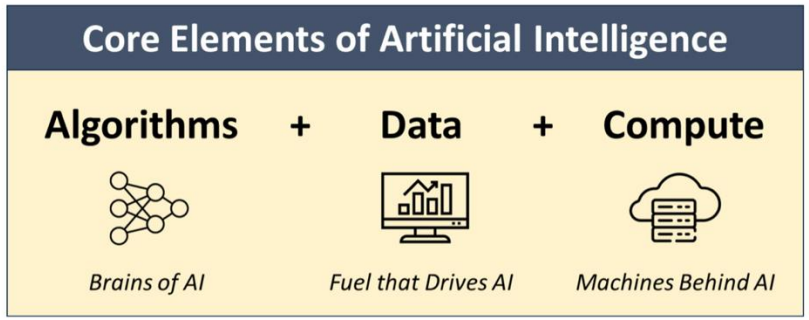


X → F → Y

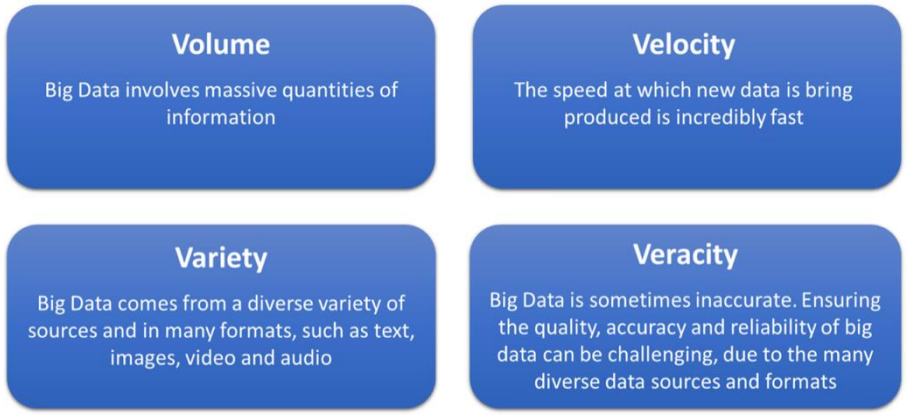
- X — Input: images, text, audio, measurements
- F — Model: learns relationships from data
- Y — Output: labels, scores, content, actions

Training adjusts F to reduce error on examples; evaluation tests F on previously unseen data.

More than data: AI requires algorithms + data + compute



The core trio



The four Vs of big data

01 Representative

Reflect the target population and use context.

02 Quality

Reliable labels, measurements, and preprocessing.

03 Governance

Permission to use, security, and source traceability.

More data is not automatically better; biased data can produce biased decisions.

Humans learn from a few experiences; models often need many examples





HUMANS

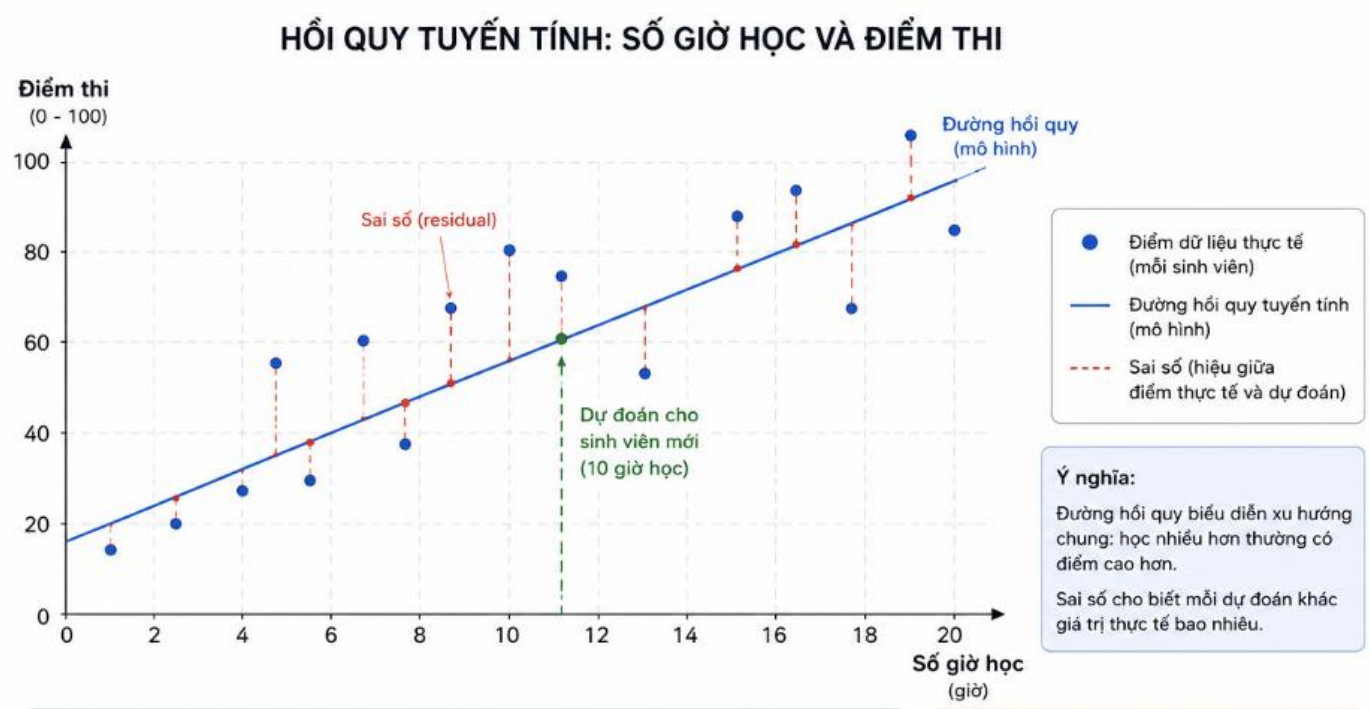
Combine observation, explanation, documentation, and prior knowledge.

MODEL

Learn relationships from many labeled examples; quality depends on diversity and label accuracy.

Shared weakness: unrepresentative examples can lead both humans and machines to generalize incorrectly.

A visual example: learning a prediction line



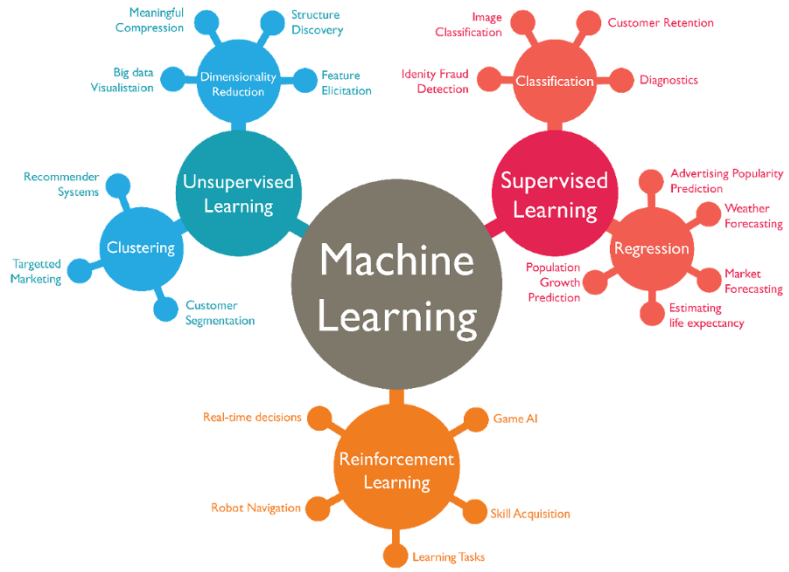
Process

1. Observe (X, Y) pairs
2. Choose a simple function family
3. Fit parameters to reduce error
4. Predict Y for a new X
5. Evaluate on unseen data

A good model does not merely memorize old points; it generalizes to new ones.

Three common learning paradigms

They differ mainly in the feedback available to the model



A visual map

SUPERVISED

Labeled data

Learn from input–output pairs.

UNSUPERVISED

Unlabeled data

Discover structure, clusters, or representations.

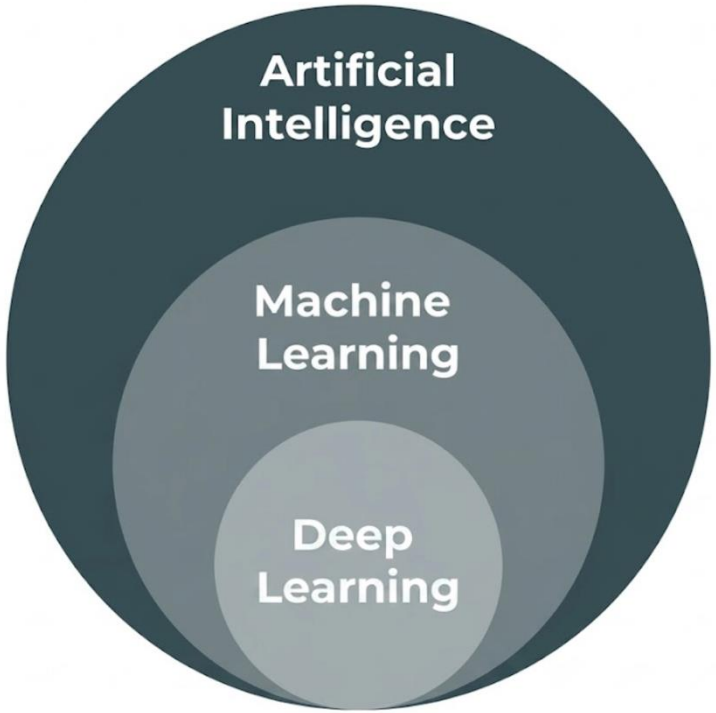
REINFORCEMENT

Reward

Try actions and optimize long-term reward.

Diagnostic question: do we have correct answers, only raw data, or only rewards for actions?

AI, Machine Learning, and Deep Learning: a nested relationship



- AI — broad scope: methods that produce intelligent behavior, including rules and search.
- ML — a branch of AI: models learn relationships from data.
- DL — a branch of ML: multilayer neural networks learn complex representations.

AI ⊃ ML ⊃ DL

Not all AI is ML; not all ML is Deep Learning.

04

How should we use AI?

Where can AI support learning and daily life — and what responsibilities must humans retain?

SECTION BREAK

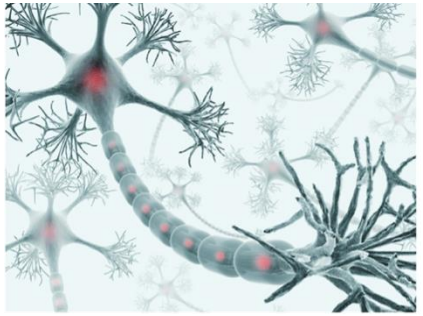
Four AI cases close to daily life and university



EDUCATION

Advising chatbot / learning assistant

Explain, suggest, and provide feedback — not complete the work for students.



HEALTHCARE

Explainable AI

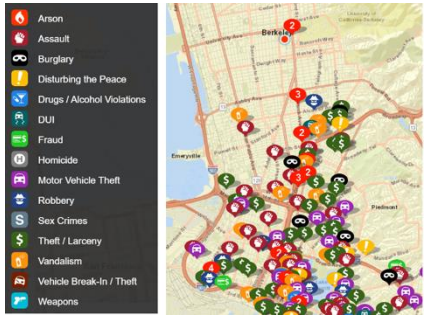
Support anomaly detection; clinicians still make decisions.



RETAIL

Product detection

Inventory products and detect missing or misplaced items.



URBAN SYSTEMS

Event-location analysis

Identify spatial patterns to support resource allocation.

Common pattern: AI suggests / detects / predicts → a human checks → a contextual decision is made.

PRACTICE

Responsible AI use: a verification loop

01

Define the objective

Do you need ideas, explanations, or evidence?

02

Limit the data

Do not enter personal or confidential data without permission.

03

Structure the request

State the context, criteria, and output format.

04

Verification

Cross-check sources, calculations, and important claims.

05

Take responsibility

Edit, cite, and own the final decision.

Hallucination

Plausible-sounding but false information

Bias

Data bias becomes output bias

Privacy

Exposure of personal or organizational data

Overreliance

Skill loss and excessive delegation of decisions

Five ideas to take away

1	**AI is a set of capabilities**	It is not a complete copy of human intelligence.
2	**Data replaces some hand-written rules**	Models learn patterns but depend on their examples.
3	**Applications are highly diverse**	Images, language, generation, signals, robotics, and Edge AI.
4	**AI ⊃ ML ⊃ DL**	This is a nested relationship, not three synonyms.
5	**Humans retain final responsibility**	Verification, security, attribution, and decisions.

QUICK CHECK

A chatbot gives a very fluent answer. What does that demonstrate?

A. It always understands correctly

B. It always has reliable sources

C. It generates language that fits the context

D. It has human-like awareness

Answer: C — fluency is a capability; correctness still requires verification.

Closing question: in your field, which tasks should AI support — and which must keep a human in the loop?